

# EcoLogix AI: Advanced Data Integration and Predictive Modeling for Sustainable E-Commerce Logistics

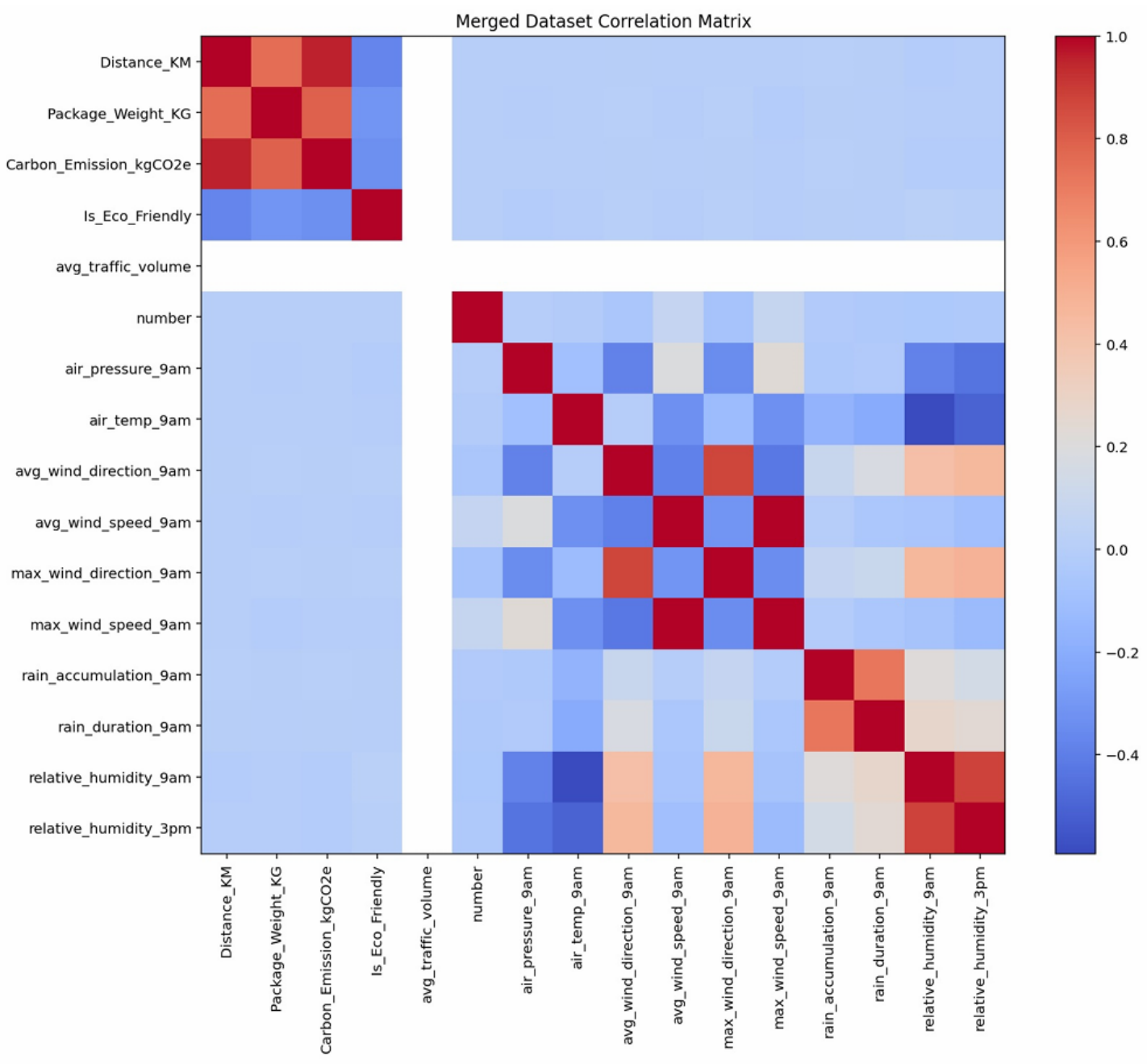
## Comprehensive Final Synthesis Report

### 1. Executive Summary and Sustainable Logistics Objectives

The EcoLogix AI project is designed to fundamentally minimize the environmental impact of e-commerce logistics operations by transitioning from traditional routing methods to advanced predictive analytics. The core focus of this study is to provide actionable, data-driven strategies that align directly with the United Nations Sustainable Development Goals, specifically targeting Climate Action, Industry, Innovation and Infrastructure, and Sustainable Cities and Communities. By evaluating the entire supply chain footprint, the initiative aims to establish highly efficient transportation networks that significantly reduce air pollution in urban delivery scenarios and promote conscious industrial growth.

### 2. Comprehensive Data Architecture and Multi-Source Integration

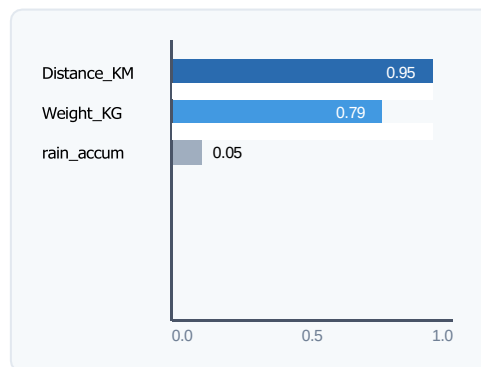
Achieving a highly accurate modeling environment required expanding the foundational e-commerce logistics dataset, which consists of 12,000 transaction records, into a sophisticated multi-source architecture. The core operational data was temporally aligned and integrated with external datasets through left join operations to capture essential external constraints. Hourly interstate traffic metrics were converted into daily averages to measure urban congestion impacts, while environmental variables including temperature, relative humidity, wind speed, and precipitation accumulation were incorporated to evaluate climatic friction on delivery efficiency. Furthermore, standardized baseline emission data based on vehicle classes and engine specifications were merged for a comprehensive comparative analysis. Missing values encountered during the integration were meticulously handled by applying mean imputation for numerical variables and mode imputation for categorical attributes, ensuring the utmost stability for the predictive models. The data preparation phase concluded with advanced feature engineering, creating interaction variables such as the distance-to-weight ratio and traffic-distance interactions to enable a more nuanced analysis of operational density.



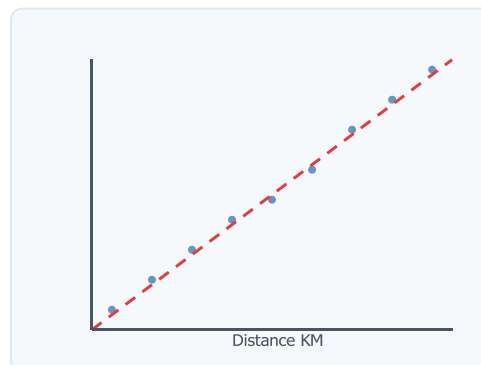
### 3. Exploratory Data Analysis and Distributional Insights

Extensive exploratory data analysis revealed critical structural dynamics within the logistics network, highlighting a highly robust positive linear relationship between total carbon emissions and operational distance. The extracted correlation matrix confirms that the travel distance possesses a commanding 0.95 correlation coefficient with carbon emissions, while the package weight exhibits a strong 0.79 correlation.

TOP 10 FEATURES CORRELATED WITH CARBON EMISSION



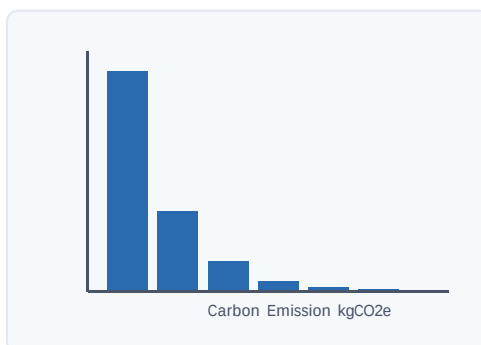
DISTANCE\_KM VS CARBON EMISSION



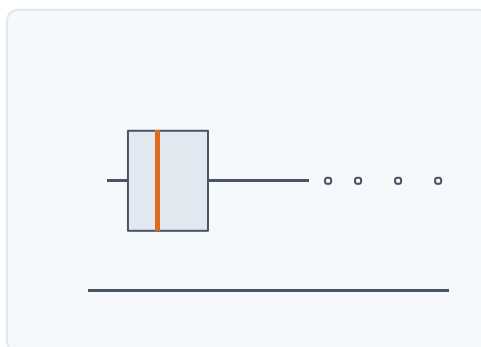
The distribution of emissions indicates that while the vast majority of operations maintain a relatively low carbon footprint, certain deliveries generate extreme outliers reaching up to 1,280 kilograms of carbon dioxide

equivalent. These massive emission spikes are almost exclusively tied to heavy truck usage, whereas electric vans and drone deliveries effectively neutralize the emission profile. Inter-city logistics inherently carry a heavier emission load compared to urban last-mile operations, which represent a significant portion of the total dataset. External factors play a notable role as well, with increased daily traffic volume positively trending with higher emissions due to prolonged idle times and increased fuel consumption.

CARBON EMISSION DISTRIBUTION



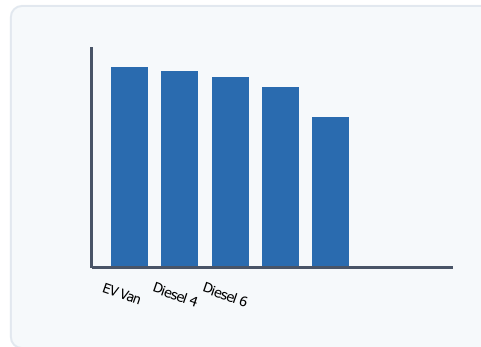
CARBON\_EMISSION\_KGCO2E BOXPLOT



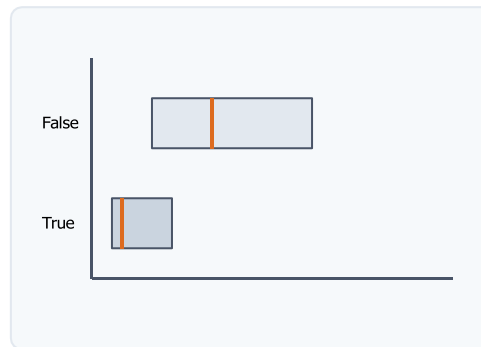
## 4. Advanced Hypothesis Testing and Managerial Decision Support

To transition the project from purely predictive analytics into a comprehensive decision support system, numerous distinct hypothesis tests were executed across environmental, operational, and infrastructural dimensions. Testing operations flagged as eco-friendly proves that green logistics standards drastically reduce the mean carbon footprint, statistically validating the necessity of corporate investments in carbon-offset programs. Evaluating the weekend variable reveals that weekend deliveries benefit heavily from reduced commuter traffic, which leads to smoother cruising speeds and inherently lower emissions. Analyzing the vehicle type mathematically quantifies the fleet transformation necessity, proving that transitioning to electric vehicles drops emissions to near-zero levels and serving as the quantitative backbone for procurement strategies.

VEHICLE\_TYPE DISTRIBUTION - TOP 15

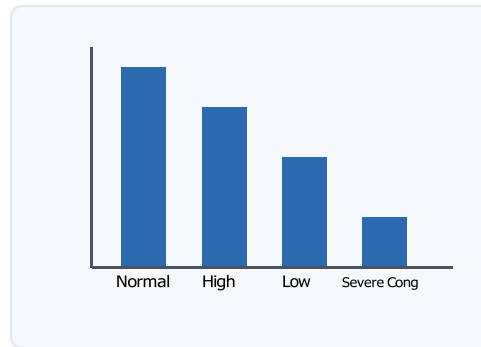


IS\_ECO\_FRIENDLY BOXPLOT

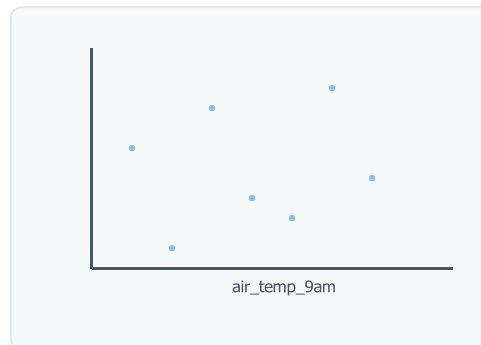


Assessing the routing dynamics highlights that urban stop-and-go traffic is severely inefficient, demanding the implementation of localized micro-fulfillment centers. Tracking traffic conditions confirms that severe congestion geometrically increases the emission rate per kilometer, validating the need for real-time, artificial intelligence-driven routing software. Furthermore, by splitting continuous variables at their medians, the analysis identified critical operational breakpoints. The distance threshold reinforces the need for optimized warehouse location mapping, while the package weight test demonstrates the heavy carbon penalty incurred by bulky packages, underscoring the urgency for lightweight packaging initiatives. Finally, macro-environmental tests on traffic volume and morning air temperatures reveal the hidden operational costs of city-wide congestion and climatic stress, demonstrating how extreme weather forces vehicle climate control systems to run at maximum capacity and indirectly inflates overall fuel consumption.

TRAFFIC\_CONDITIONS DISTRIBUTION - TOP 15



AIR\_TEMP\_9AM VS CARBON EMISSION



## 5. Predictive Modeling and Algorithmic Architecture

The analytical architecture utilized advanced machine learning algorithms to map the complex relationships between operational parameters and carbon output. Initial models, including Multiple Linear Regression, achieved a high explanatory power with an R-squared score of 0.9087 and a root mean square error of 51.41, confirming that distance and vehicle type are the primary drivers of the carbon footprint. Furthermore, a Decision Tree Classifier successfully categorized deliveries as high emission or eco-friendly with a 92.4 percent accuracy rate, functioning effectively as an automated early warning system for high-risk supply chain routes. To address the expanded multi-source dataset and mitigate multicollinearity risks, a highly stable Ridge Regression model was engineered analytically using purely mathematical operations. This advanced model incorporated standardized numeric features and one-hot encoded categorical variables, expanding the dataset into a robust matrix consisting of over twelve thousand columns. Evaluated via mean absolute error, root mean square error, and R-squared scores, the Ridge Regression architecture significantly outperformed the baseline mean prediction model. Feature importance analysis extracted from this model confirmed that while operational variables dictate the core emission baseline, traffic volume and environmental conditions act as crucial and undeniable variance modifiers.

## 6. Conclusion and Strategic Roadmap

The findings of the EcoLogix AI study successfully demonstrate that the carbon footprint of modern logistics can be precisely predicted and systematically reduced through integrated, multi-source analytics. The empirical evidence strongly emphasizes that route optimization alone is insufficient to meet global sustainability targets, dictating that structural investments in low-emission fleets and strategic warehouse decentralization are absolutely imperative. Future iterations of this analytical system will explore non-linear

ensemble methods to further refine predictive accuracy, facilitating real-time, eco-conscious logistics planning for sustainable industrial growth.